

RAMKUMAR U

Full Stack Developer

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SUMMARY

Full Stack Developer with 2+ years of experience building production web, desktop, and mobile applications using React (TypeScript), Node.js, Python (Flask/FastAPI), and MySQL. Experienced in owning products end-to-end — frontend UI, backend APIs, databases, third-party integrations (payment gateways, WhatsApp, IndiaMART), applied AI (RAG-based LLM assistants), and AWS/Firebase deployment. Strong problem-solver focused on building scalable, secure, and user-centric software.

CORE TECHNICAL SKILLS

- Frontend:** React.js, Redux, Context API (Global State Management), TypeScript, Sass/SCSS, MUI, Bootstrap 5
- Backend:** Node.js, Express.js, Python (Flask, FastAPI), REST API design, JWT Auth
- Database:** MySQL — schema design, query optimization, data handling
- Cloud & Deployment:** AWS EC2 (provisioning, security groups, firewall, reverse proxy, domain/SSL setup), AWS Amplify, Firebase Hosting, Windows IIS deployment
- Desktop & Mobile:** Electron (auto-launch, auto-updater, logging) for Windows desktop apps; Capacitor for Android (APK) apps
- AI & Integrations:** RAG-based LLM knowledge assistants, Meta WhatsApp Business API, IndiaMART API, PhonePe Payment Gateway, SSO (Microsoft account), third-party lead/CRM integrations
- Tools, Methodologies & Practices:** Git/GitHub, Postman, Agile/Scrum, Jira, pandas, numpy

EXPERIENCE

- Full Stack Developer — Bharath Vending Corporation** Feb 2024 – Present
Full Stack Developer — ERP, CRM & CMS Platforms
- End-to-end ownership:** designed and delivered web dashboards, Windows desktop apps, and Android apps to manage business operations and vending machine networks.
 - Backend & architecture:** built backend architecture, MySQL database design, and REST APIs powering ERP, CRM, and CMS platforms.
 - Cloud deployment:** managed production servers on AWS EC2, including security groups, firewall configuration, reverse proxy, and domain setup.
 - Payments & integrations:** integrated PhonePe payment gateway, Meta WhatsApp Business API, and IndiaMART lead capture across multiple internal platforms.
- Full Stack Developer — Seilax Software Systems Pvt. Ltd.** Sep 2023 – Feb 2024
Innovative Web Solutions
- Shipped production apps:** Company Profile Website, Real Estate Portal, and Mall Leasing & Management System using React, Node.js, and Firebase.
 - Frontend architecture:** built scalable, reusable React component architecture with centralized API services and global state management (MUI, Bootstrap, Framer Motion, Swiper, Axios, React Toastify, XLSX).
 - Backend APIs:** developed secure RESTful APIs with Node.js/Express (Axios, Body Parser, CORS, dotenv, Nodemon).
 - Live projects:** [www.seilaxsoftware.com](#) | [portal.seilaxsoftware.com](#)

PROJECTS

- ERP Software with AI Assistant — Bharath Vending Corporation**
- Designed and built a fully automated, company-wide ERP system from scratch covering operations, inventory, and lead management.
 - Integrated a RAG-based LLM AI assistant trained on internal company knowledge to answer employee/business queries instantly.
 - Built Meta WhatsApp Business API integration for automated customer communication directly from the ERP.
 - Connected IndiaMART API to automatically capture and sync inbound sales leads into the ERP in real time.
- Vending Machine CRM — Bharath Vending Corporation**
- Built a CRM to manage vending machine operations with payment-based transaction reporting for full financial visibility.
 - Integrated PhonePe payment gateway for seamless cashless transactions across machines.
 - Enabled remote monitoring and testing of machines from anywhere through the CRM dashboard.
 - Implemented real-time stock monitoring and role-based access control for secure, permission-based operations.
- Decal Vending Machine CMS — Caterpillar**
- Developed a CMS with job ID and department-based tracking for organized, traceable operations.
 - Built role-based reporting so managers and staff see reports scoped to their role and department.
 - Enabled remote monitoring and testing of vending machines directly from the CMS.
 - Added Excel bulk upload support for fast, large-scale data entry and updates.
 - Designed a robust RBAC (Role-Based Access Control) system for both roles and individual employees.
 - Integrated SSO (Single Sign-On) with Microsoft accounts for secure, seamless user authentication.

EDUCATION

- BE in Mechatronics Engineering** 2023
M.A.M School of Engineering, Trichy, India